

HTML (Advanced)

1. Create a semantic blog page layout using <header>, <nav>, <article>, <aside>, and <footer> with dummy content.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>RGA Blog Page</title>
  <style>
    *{box-sizing:border-box}
    body{
      margin:0;
      font:16px/1.6 system-ui, Arial, sans-serif;
      color:#1f2937;
      background:#fafafa;
    }
    a{color:#2563eb;text-decoration:none}
    a:hover{text-decoration:underline}
    .wrap{max-width:1180px;margin:0 auto;padding:0 16px}
    header{
      background:linear-gradient(150deg,#e3fbda,#fafff2);
      color:#0f172a;
    }
    header .wrap{padding:40px 16px;text-align:center}
    h1{margin:0 0 8px;font-size:36px}
    .tagline{margin:0;font-size:18px;font-weight:500}
    nav{
      background:linear-gradient(120deg,#d4fc79,#96e6a1);
    }
    .nav-inner{display:flex;justify-content:center;gap:24px;padding:12px 0}
    nav a{font-weight:600;color:#064e3b}
    .content{
      display:grid;
      grid-template-columns: 2fr 1fr;
      gap:20px;
      padding:20px 0;
    }
    article{
      background:linear-gradient(120deg,#e3fbda,#fafff2);
      border-radius:12px;
      padding:22px;
      margin-bottom:20px;
      box-shadow:0 4px 10px rgba(0,0,0,0.05);
    }
    article h2{margin:6px 0 14px;font-size:26px}
    .kicker{display:inline-block;font-size:13px;font-weight:700;letter-spacing:.05em;text-transform:uppercase;color:#9333ea}
    aside{
      background:linear-gradient(120deg,#e3fbda,#fafff2);
      border-radius:12px;
      padding:20px;
      box-shadow:0 4px 12px rgba(0,0,0,0.06);
      height:fit-content;
    }
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}
aside h3{margin:6px 0 12px;color:#7c2d12}
.list{margin:0;padding:0;list-style:none}
.list li{margin:10px 0}
.meta{margin-top:20px;font-size:14px;color:#444}

footer{
  margin-top:20px;
  background:linear-gradient(120deg,#e0fed2,#f6fff2);
  color:#0f172a;
  text-align:center;
}
footer .wrap{padding:20px 16px}
.credit{margin:0;font-size:15px;font-weight:500}

@media (max-width: 820px){
  .content{grid-template-columns:1fr}
}
</style>
</head>
<body>
  <header>
    <div class="wrap">
      <h1>Welcome to RGA Blog Page</h1>
      <p class="tagline">A space where thoughts turn into stories and stories
inspire lives.</p>
    </div>
  </header>
  <nav>
    <div class="wrap nav-inner">
      <a href="#">Home</a>
      <a href="#">Journey</a>
      <a href="#">Connect</a>
    </div>
  </nav>
  <main class="wrap content" role="main">
    <div>
      <article>
        <span class="kicker">Lifestyle</span>
        <h2>Chasing Sunsets: Why Evenings Teach Us Balance</h2>
        <p><strong>Introduction –</strong> Every sunset tells a story. It's
not just the end of the day—it's a reminder that endings can be just as
beautiful as beginnings. In a world that moves too fast, sunsets invite us to
slow down and breathe.</p>
        <p><strong>A Pause in Time.</strong> Sunsets prove stillness can be
stunning. The sky slows, and so do our thoughts.</p>
        <p><em>Sometimes, the best therapy is simply watching the sky change
colors. </em></p>
      </article>

      <article>
        <span class="kicker">Well-being</span>
        <h2>The Joy of Rainy Days</h2>
        <p>Rainy days carry a magic of their own. The rhythm of raindrops
creates a soundtrack for calmness and reflection, while cozy corners and warm
drinks become our favorite companions.</p>
        <p><em>A rainy day is not wasted; it's a reminder to embrace the

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quiet. </em></p>
</article>
<article>
  <span class="kicker">Minimalism</span>
  <h2>Living More with Less</h2>
  <p>Minimalism is about making space for what truly matters. When we
let go of clutter, we create room for clarity, creativity, and calm.</p>
  <p><em>Minimalism is not subtraction, it's multiplication of peace.
</em></p>
</article>
</div>
<aside aria-label="Sidebar">
  <h3>Popular Reads</h3>
  <ul class="list">
    <li> <a href="#">The Art of Doing Nothing</a></li>
    <li> <a href="#">Conversations Over Coffee</a></li>
    <li> <a href="#">Playlists That Heal the Soul</a></li>
  </ul>
  <div class="meta">
    <h3>Post Info</h3>
    <div><strong>Last Updated:</strong> August 29, 2025</div>
    <div><strong>Author:</strong> RGA Creative Corner</div>
  </div>
</aside>
</main>
<footer>
  <div class="wrap">
    <p class="credit">© 2025 <strong>RGA Blog</strong> – Made with stories,
sunsets, and coffee</p>
  </div>
</footer>
</body>
</html>

```

2. Implement an HTML5 video player with controls, autoplay, muted, and fallback text if the browser doesn't support <video>.

```

<!DOCTYPE html>
<html>
<head>
  <title>Video Example</title>
</head>
<body>
  <h2>JOG FALLS</h2>
  <video width="900" controls autoplay muted>
    <source src="jog falls.mp4" type="video/mp4">
    Sorry, your browser doesn't support HTML5 video.
  </video>
</body>
</html>

```

3. Build an HTML form that includes validation rules (required fields, email format, number range between 1–100).

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
</head>
<body>
  <h2>Form</h2>
  <form id="myForm">
    Name: <input type="text" required><br><br>
    Email: <input type="email" required><br><br>
    Age (1-100): <input type="number" min="1" max="100" required><br><br>
    <input type="submit" value="Submit">
  </form>
  <script>
    document.getElementById("myForm").addEventListener("submit",
function(event) {
    event.preventDefault();
    alert("Form submitted successfully!");
  });
  </script>
</body>
</html>
```

CSS (Advanced)

4. Using CSS Grid, create a 3x3 grid layout where the middle cell spans across 2 columns.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>3 X 3 GRID</title>
</head>
<style>
  .grid {
    display: grid;
    grid-template-columns: repeat(3, 100px);
    grid-template-rows: repeat(3, 100px);
    gap: 5px;
  }
  .grid div {
    background: lightblue;
    display: flex;
    justify-content: center;
    align-items: center;
    font-size: 20px;
    font-weight: bold;
  }
  .middle {
    grid-column: 2 / 4;
    background: coral;
  }
```

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    }
  </style>
</head>
<body>
  <h2>3x3 Grid Layout</h2>
  <div class="grid">
    <div>${</div>
    <div>${</div>
    <div>${</div>
    <div>${</div>
    <div class="middle">${</div>
    <div>${</div>
    <div>${</div>
    <div>${</div>
  </div>
</body>
</html>

```

5. Design a responsive navigation bar using only CSS (without JavaScript) that turns into a hamburger menu on small screens.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Navbar</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
    }
    nav {
      background: #333;
      color: white;
      padding: 10px;
    }
    nav ul {
      list-style: none;
      margin: 0;
      padding: 0;
      display: flex;
    }
    nav ul li {
      margin: 0 15px;
    }
    nav ul li a {
      text-decoration: none;
      color: white;
      font-weight: bold;
    }
    .menu-toggle {
      display: none;
      flex-direction: column;
      cursor: pointer;

```

```

    }
    .menu-toggle span {
      height: 3px;
      width: 25px;
      background: white;
      margin: 4px 0;
      border-radius: 2px;
    }
    @media (max-width: 600px) {
      nav ul {
        display: none;
        flex-direction: column;
        background: #333;
      }
      nav input[type="checkbox"]:checked ~ ul {
        display: flex;
      }
      .menu-toggle {
        display: flex;
      }
    }
  }
</style>
</head>
<body>
  <nav>
    <input type="checkbox" id="menu-toggle" style="display:none;">
    <label for="menu-toggle" class="menu-toggle">
      <span></span>
      <span></span>
      <span></span>
    </label>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">About</a></li>
      <li><a href="#">Services</a></li>
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>
</body>
</html>

```

6. Create a CSS animation where a circle moves left to right infinitely across the screen.

```

<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
  <title>Moving Circle Animation</title>
  <style>
    .circle {
      width: 50px; height: 50px;
      background: purple; border-radius: 50%;
      position: absolute;
      animation: move 3s linear infinite;
      align-items: center;
    }
  </style>
</head>
<body>
  <div>
    <div class="circle"></div>
  </div>
</body>
</html>

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        justify-content: center;
    }
    @keyframes move {
        from { left: 0; }
        to { left: 90%; }
    }
</style>
</head>
<body>
    <div class="circle"></div>
</body>
</html>

```

JavaScript (Advanced)

- Write a JavaScript program to implement debouncing for a search input field (only trigger API call when user stops typing for 500ms).

```

<!DOCTYPE html>
<html>
<body>
    <input type="text" id="search" placeholder="Type to search...">
    <p id="output"></p>
    <script>
        let timer;
        document.getElementById("search").addEventListener("input", function(e) {
            clearTimeout(timer);
            timer = setTimeout(() => {
                document.getElementById("output").innerText = "Searching for: " +
e.target.value;
            }, 500);
        });
    </script>
</body>
</html>

```

- Implement a drag and drop feature where a user can drag an image into a drop zone.

```

<!DOCTYPE html>
<html>
<body>
    <h3>Drag an image</h3>
    
    <div id="dropzone"
        style="width:300px;height:200px;border:2px dashed black;margin-
top:20px;text-align:center;line-height:200px;"
        ondrop="drop(event)"
        ondragover="allowDrop(event)">
        Drop Here
    </div>
    <script>
        function allowDrop(ev) {

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        ev.preventDefault();
    }
    function drag(ev) {
        ev.dataTransfer.setData("id", ev.target.id);
    }
    function drop(ev) {
        ev.preventDefault();
        var data = ev.dataTransfer.getData("id");
        var dragged = document.getElementById(data);
        ev.target.innerHTML = "";
        ev.target.appendChild(dragged);
    }
</script>
</body>
</html>

```

9. Write a JavaScript function that fetches data from a public API (e.g., JSONPlaceholder) and dynamically displays it inside a `<ul>` list.

```

<!DOCTYPE html>
<html>
<body>
    <h3>Users List</h3>
    <ul id="list"></ul>
    <script>
        fetch("https://jsonplaceholder.typicode.com/users")
            .then(function(res){
                return res.json();
            })
            .then(function(data){
                var ul = document.getElementById("list");
                for(var i=0; i<data.length; i++){
                    var li = document.createElement("li");
                    li.innerHTML = data[i].name;
                    ul.appendChild(li);
                }
            });
    </script>
</body>
</html>

```

10. Create a to-do list app in JavaScript where users can add tasks, mark them as completed, and delete them.

```
<!DOCTYPE html>
<html>
<body>
  <h2>To Do List</h2>
  <input id="task" placeholder="Add task">
  <button onclick="add()">Add</button>
  <ul id="list"></ul>
  <script>
    function add(){
      var t = document.getElementById("task").value;
      if(t==""){
        alert("Enter task");
        return;
      }
      var li = document.createElement("li");
      li.innerHTML = t + " ";
      var cbtn = document.createElement("button");
      cbtn.innerHTML = "Complete";
      cbtn.onclick = function(){
        li.style.color = "green";
        cbtn.innerHTML = "Done";
      }
      var dbtn = document.createElement("button");
      dbtn.innerHTML = "Delete";
      dbtn.onclick = function(){
        li.remove();
      }
      li.appendChild(cbtn);
      li.appendChild(dbtn);
      document.getElementById("list").appendChild(li);
      document.getElementById("task").value = "";
    }
  </script>
</body>
</html>
```